Needs-based recommendation systems

Machine Learning Pipeline for Financial Needs Prediction and Product Recommendation

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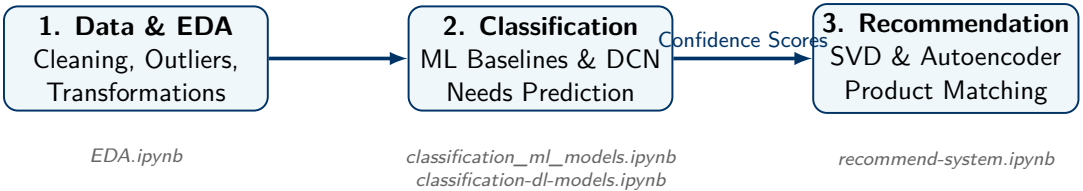
Project Overview



Data Quality Report

- **Missing Values & Cleaning:** The initial dataset was largely complete. Invalid categorical codes (e.g., `Gender=0`) were mapped to missing and imputed. There were no zero or near-zero variance columns.
- **Variable Distributions:**
  - *Symmetric/Normal:* `Age`, `FinancialEducation`, and `RiskPropensity` follow near-normal distributions.
  - *Highly Skewed:* Financial variables (`Wealth`, `Income`) exhibit extreme right-skewness and heavy tails, reflecting structural economic inequality.
- **Outlier Handling:** Isolation Forest flagged outliers, but extreme financial values are valid business profiles, not errors. Consequently, robust scaling and **logarithmic transformations** were favored over outright deletion to stabilize variance.

Key EDA Findings

- **Feature Predictive Power:** Violin plots reveal individuals needing investments are older, wealthier, and exhibit higher income and risk propensity. Conversely, `Gender` and `FamilyMembers` showed minimal discriminative power.
- **Target Independence:** A Chi-squared (χ^2) test confirmed that *Income Investment* and *Accumulation Investment* needs are statistically independent, validating a multi-label/multi-task modeling approach.
- **Target Class Imbalance:**
  - *Accumulation Investment* is perfectly balanced ($\sim$50% prevalence).
  - *Income Investment* exhibits moderate imbalance ($\sim$30% prevalence).
  - *Conclusion:* Accuracy is an unreliable metric, justifying the optimization of the **F0.5 Score** in the modeling phase.

Feature Transformation and Engineering

1. Data Transformation (`DataTransformer`)

- **Numeric Conversion:** Coercion of mixed/string formats to continuous variables.
- **Log Transformation:** Applied $\log(1 + x)$ to highly skewed financial variables (`Wealth`, `Income`) to normalize distributions.
- **Scaling:** `StandardScaler` applied post-engineering for stable convergence.

2. Household Metrics

- `WealthIncomeRatio`
- `IncomePerFamilyMember`

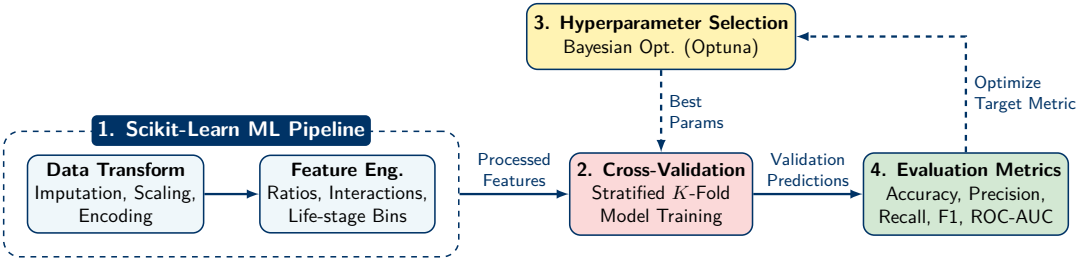
3. Interaction Features

- `RiskEducationInteraction` ($Risk \times Edu$)
- `RiskWealthInteraction` ($Risk \times \log(Wealth)$)
- `AgeRiskInteraction` ($Age \times Risk$)

4. Non-linearity & Life-Stages

- `AgeSquared`: Models the "Life-Cycle" curve (wealth peaking in mid-life).
- **Age Bins**: One-hot flags for `Under35`, `35_54`, `55_69`, and `70plus`.

ML Pipeline Overview



Baseline Models Evaluated

Models Evaluated

We comprehensively tested several families of classifiers across both target variables (*Income* and *Accumulation*):

- **Distance/Linear**: SVM, Naive Bayes, KNN.
- **Tree-based Ensembles**: Random Forest, XGBoost, LightGBM.
- **Meta-Ensembles**:
  - Voting Classifier (Soft & Hard)
  - Stacking Classifier (LGBM, XGB, KNN, RF with a Logistic Regression meta-learner)

Top Performers & Selection

Selected for Tuning:

- **Random Forest**
- **LightGBM**

Selection Rationale:

- Strong baseline predictive power with robust handling of non-linear interactions.
- **Target Metric:** Optimised for **F0.5 Score** to prioritise precision over recall (minimising the risk of unsuitable, false-positive product recommendations).

Optuna Hyperparameter Tuning

Objective Metric

- Optimized for **F0.5 Score** (Stratified 5-Fold CV)
- *Rationale:* Prioritizes precision over recall ($\beta = 0.5$), significantly penalizing false positives (bad recommendations).

TPE Sampler Strategy

- Uses Tree-structured Parzen Estimator (TPE).
- Explores the search space efficiently by modelling the probability of trial performance.

Hyperparameter Search Space

Random Forest (20 Trials)

- `n_estimators`: 10–200
- `max_depth`: 3–10
- `min_samples_split/leaf`: Constraints for regularization.

LightGBM (50 Trials)

- `learning_rate`: 0.001–0.2 (Log scale)
- `num_leaves`: 15–255
- `reg_alpha` & `reg_lambda`: L1/L2 Regularization (Log scale)

Optimization Results

- LightGBM consistently outperformed Random Forest across both prediction targets.
- The tuned models demonstrated a significant reduction in false positives while maintaining competitive accuracy.

	Income Target			Accumulation Target		
Model	**Precision**	**Recall**	**F0.5 Score**	**Precision**	**Recall**	**F0.5 Score**
Baseline RF	0.724	0.450	0.648	0.763	0.630	0.733
Tuned RF	**0.785**	0.412	0.669	**0.801**	0.582	0.745
Baseline LGBM	0.760	0.521	0.696	0.812	0.685	0.783
Tuned LGBM	**0.802**	0.505	**0.720**	**0.845**	0.671	**0.803**

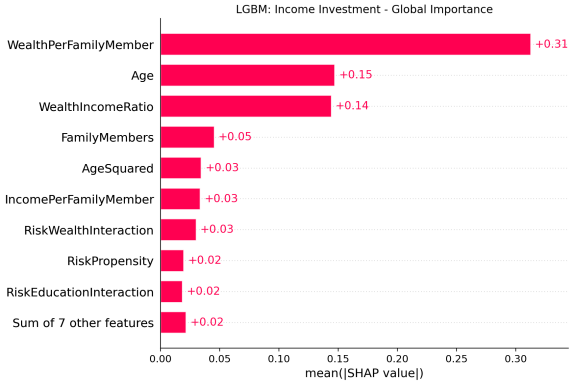
Final Selection

Tuned LightGBM was selected as the final model for generating probabilities to feed into the recommendation engine, ensuring highly precise product targeting.

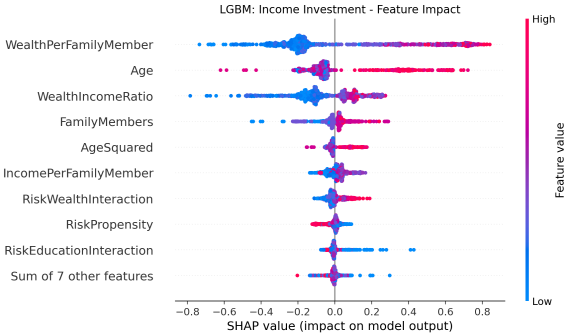
XAI Methods: SHAP, LIME & Permutation

- **SHAP (SHapley Additive exPlanations):** Used to understand global feature importance and directionality across the entire dataset. It highlights which engineered features drive predictions the most.
- **LIME (Local Interpretable Model-agnostic Explanations):** Used to explain individual, client-level predictions to build trust and provide clear reasoning for a recommended product.

Income Investment Model Insights (LightGBM)

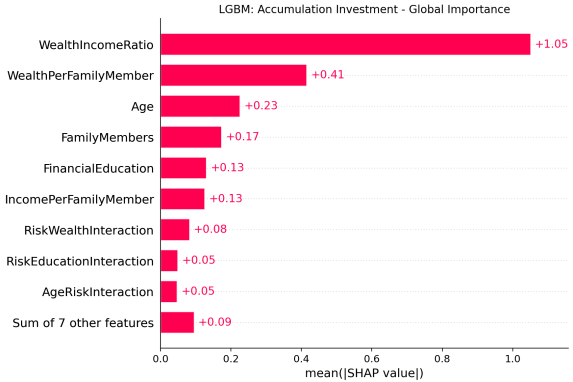


SHAP Feature Importance

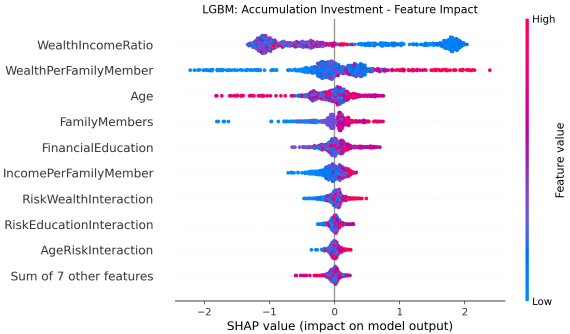


SHAP Beeswarm Plot

Accumulation Investment Model Insights (LightGBM)

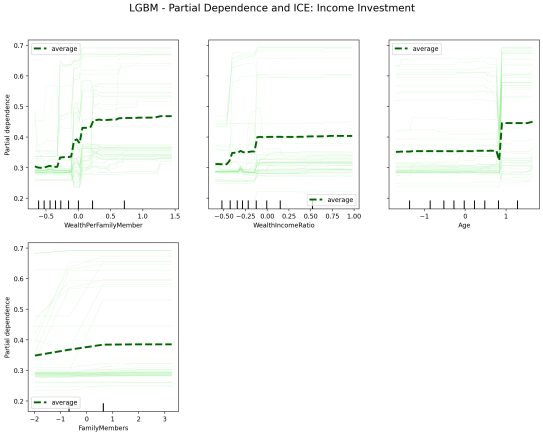


SHAP Feature Importance



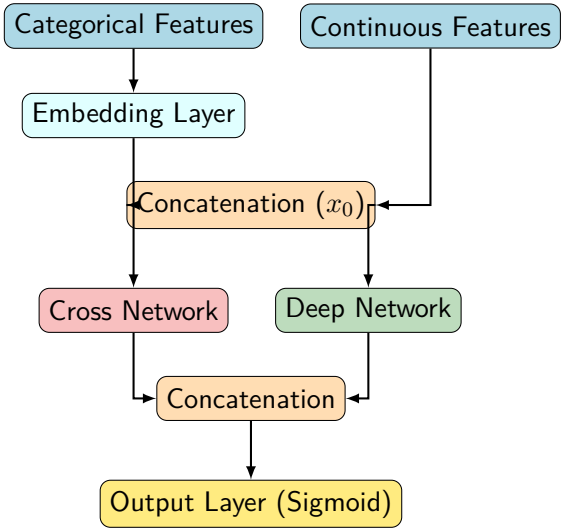
SHAP Beeswarm Plot

LIME: Local Interpretability Example



Example LIME Explanation for a single client prediction (LightGBM Income Model)

Deep Learning Model Architecture: DCN [1]



Multi-Task Learning Results

Task	Acc	Prec	Rec	F1	AUC
Income	0.783	0.852	0.544	0.664	0.767
Accum.	0.795	0.846	0.708	0.771	0.852

Architecture Highlights

- **Cross Network**: Applies explicit bounded-degree feature crossing efficiently.
- **Deep Network**: Learns implicit non-linear higher-order interactions.
- **Joint Training**: Both networks share the same concatenated input embeddings, optimising for dual investment targets.

Methodology: User-Item Matrix

We construct a user-item interaction matrix R using predicted probabilities from the supervised classifiers, penalised by risk misalignment.

Matrix Entry R_{ij}

$$R_{ij} = P_{\text{type}(j)}(i) \cdot \big(1 - \alpha \cdot |\text{RiskPropensity}(i) - \text{Risk}(j)|\big)$$

- $P_{\text{type}(j)}(i)$: Predicted probability of client i needing product j's type (Accumulation or Income).
- $\alpha \in [0, 1]$: Decay factor penalising risk misalignment.
  - **Economic Rationale**: Reflects the firm's risk aversion and regulatory suitability constraints (e.g., MiFID II).
  - **Trade-off**: Balances maximising expected fee revenue/conversion against minimising compliance risks and long-term customer churn from unsuitable investments.

Methodology: Models & Evaluation

We use model-based CF [2] leveraging matrix R:

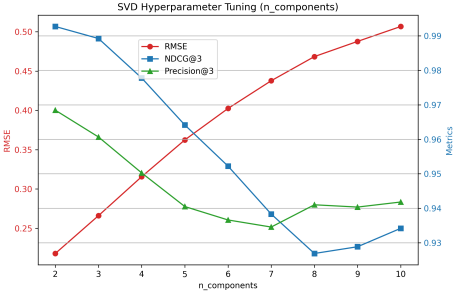
- **Truncated SVD**: Factorises $R \approx U_k \Sigma_k V_k^T$ into low-dimensional latent factors to capture hidden preferences.
- **Autoencoder**: Learns non-linear representations by compressing and reconstructing the interaction matrix ($z_u = f_\theta(r_u)$, $\hat{r}_u = g_\phi(z_u)$).

Evaluation Metrics

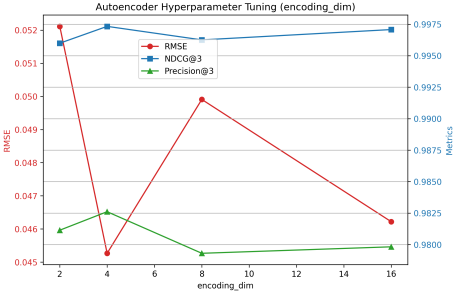
- **RMSE**: Measures the model's reconstruction error.
- **Precision@k & Recall@k**: Evaluate the relevance of the top-k recommendations.
- **NDCG@k**: Assesses ranking quality, prioritising relevant items placed higher.

Experimental Results

Model	RMSE	Precision@3	Recall@3	NDCG@3
Truncated SVD	0.402	0.937	0.937	0.952
Autoencoder	**0.039**	**0.984**	**0.984**	**0.997**



SVD Tuning



Autoencoder Tuning

Project Summary

- **Comprehensive ML Pipeline:** Integrated extensive feature engineering, missing value imputation, and log-scaling directly into a robust scikit-learn pipeline to handle skewed financial metrics.
- **Targeted Model Selection:** Selected and optimized LightGBM for the F0.5 score, successfully prioritizing precise, risk-appropriate recommendations over raw recall.
- **Deep Learning Insights:** Leveraged a Deep & Cross Network (DCN) to learn explicit feature combinations jointly across two prediction tasks.
- **Personalized Recommendations:** Built an Autoencoder-based Collaborative Filtering system that significantly outperforms SVD while enforcing business-driven risk misalignment penalties.
- **Interpretability First:** Embedded global (SHAP) and local (LIME) interpretability, ensuring that every algorithmic recommendation is explainable to financial advisors and regulatory bodies.

References

[1] Weiyu Guo et al. "Explainable enterprise credit rating using deep feature crossing". en. In: *Expert Systems with Applications* 220 (June 2023), p. 119704.

[2] Marwa Sharaf et al. "A survey on recommendation systems for financial services". In: *Multimedia Tools and Applications* 81.12 (May 2022), pp. 16761–16781.